

LPU 3D Campus Navigator

Database Schema Plan

Recommended database: PostgreSQL through Supabase. This fits the relational links between locations, vendors, facilities and the routing graph.

Entity Relationships

locations 1→many vendors; locations many↔many facilities; route_nodes 1→many route_edges; locations optionally map to route_nodes; chatbot_documents optionally reference locations; chatbot_messages belong to sessions.

locations

Column	Type	Constraints
id	UUID	PK
name	TEXT	NOT NULL
slug	TEXT	UNIQUE
type	TEXT	NOT NULL
description	TEXT	NULL
short_description	TEXT	NULL
block_code	TEXT	NULL
latitude	DOUBLE PRECISION	NULL
longitude	DOUBLE PRECISION	NULL
map_x/map_y/map_z	DOUBLE PRECISION	NULL
image_url	TEXT	NULL
model_object_id	TEXT	UNIQUE
opening_hours	JSONB	NULL
facilities	JSONB	NULL
is_active	BOOLEAN	DEFAULT TRUE
created_at/updated_at	TIMESTAMPTZ	DEFAULT NOW()

vendors

Column	Type	Constraints
id	UUID	PK
location_id	UUID	FK → locations.id
name	TEXT	NOT NULL
category	TEXT	NOT NULL
description	TEXT	NULL
phone	TEXT	NULL

Column	Type	Constraints
image_url	TEXT	NULL
opening_hours	JSONB	NULL
is_active	BOOLEAN	DEFAULT TRUE
created_at/updated_at	TIMESTAMPTZ	DEFAULT NOW()

facilities

Column	Type	Constraints
id	UUID	PK
name	TEXT	NOT NULL
description	TEXT	NULL
icon	TEXT	NULL

location_facilities

Column	Type	Constraints
location_id	UUID	FK → locations.id
facility_id	UUID	FK → facilities.id
PK	(location_id, facility_id)	Composite

route_nodes

Column	Type	Constraints
id	UUID	PK
location_id	UUID	FK → locations.id, NULL
x/y/z	DOUBLE PRECISION	NOT NULL
node_type	TEXT	NOT NULL

route_edges

Column	Type	Constraints
id	UUID	PK
from_node_id	UUID	FK → route_nodes.id
to_node_id	UUID	FK → route_nodes.id
distance_meters	DOUBLE PRECISION	NOT NULL
is_walkable	BOOLEAN	DEFAULT TRUE
accessibility	TEXT	NULL

chatbot_documents

Column	Type	Constraints
id	UUID	PK
title	TEXT	NOT NULL
content	TEXT	NOT NULL
source_type	TEXT	NOT NULL
source_url	TEXT	NULL
location_id	UUID	FK → locations.id, NULL
is_verified	BOOLEAN	DEFAULT FALSE
created_at	TIMESTAMPTZ	DEFAULT NOW()

chatbot_messages

Column	Type	Constraints
id	UUID	PK
session_id	UUID	NOT NULL
role	TEXT	NOT NULL
message	TEXT	NOT NULL
created_at	TIMESTAMPTZ	DEFAULT NOW()

Indexes

Recommended indexes: locations.name, locations.slug, locations.type, vendors.name, vendors.category, route_edges.from_node_id, route_edges.to_node_id.

Architecture Decision

The 3D model and database must share stable IDs. Example: locations.id = block-001 and the GLB object name = block-001. This connects search, map selection, chatbot responses and routing without duplicating location data.